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Set 17: Employee Performance Analysis

Aim

To analyse employee performance in R by creating a line chart of the performance trend, a bar chart of employees per department, a performance table, and a combined dashboard.

Questions (combined into one experiment)

1. Line chart of performance trend (axes labelled, titled)
2. Bar chart of employees across departments
3. Table of performance data for each employee
4. Dashboard combining the line chart, bar chart and table

Procedure

1. Open RStudio and install/load the packages ggplot2 and gridExtra.
2. Create the dataset as a data frame.
3. Draw the charts for Questions 1 and 2 using ggplot2 with titles and axis labels.
4. Display the dataset as a table using print() and tableGrob().
5. Combine the two charts and the table into a dashboard using grid.arrange().
6. Run the program and observe the output.

Code

```
library(ggplot2)
library(gridExtra)

# Dataset
df <- data.frame(EmployeeID = 1:3, Department = c("Sales", "HR",
"Marketing"),
                 YearsOfService = c(5, 3, 7), PerformanceScore = c(85, 92,
78))

# Q1. Line chart of performance trend
p1 <- ggplot(df, aes(x = YearsOfService, y = PerformanceScore)) +
  geom_line(color = "blue") + geom_point(size = 3) +
  labs(title = "Employee Performance Trend", x = "Years of Service", y =
"Performance Score")
print(p1)

# Q2. Bar chart of employees per department
p2 <- ggplot(df, aes(x = Department)) + geom_bar(fill = "orange") +
  labs(title = "Employees by Department", x = "Department", y = "Number of
Employees")
print(p2)

# Q3. Table of performance data
```

```
print(df)
tb <- tableGrob(df, rows = NULL)

# Q4. Dashboard (line chart + bar chart + table)
grid.arrange(p1, p2, tb, ncol = 2, top = "Employee Performance Dashboard")
```

Output

Table (Question 3):

```
  EmployeeID Department YearsOfService PerformanceScore
1          1      Sales              5               85
2          2         HR              3               92
3          3  Marketing              7               78
```

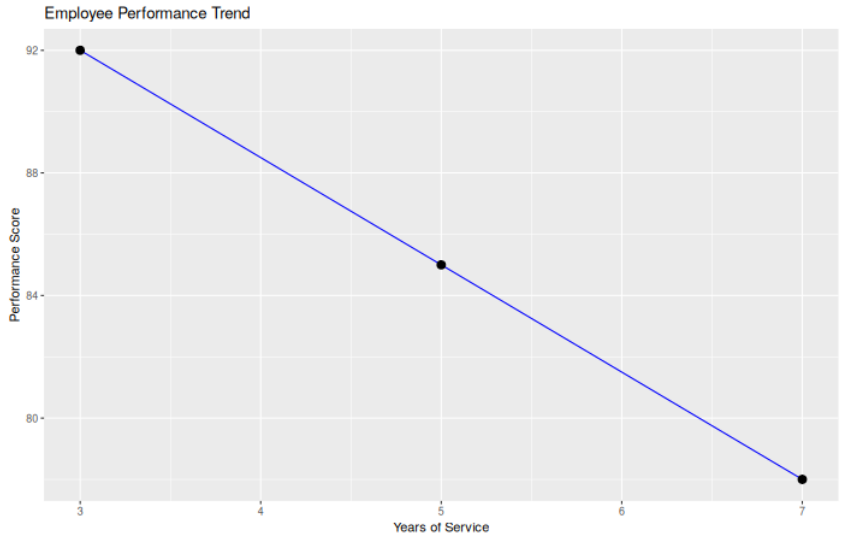


Figure 1: Question 1

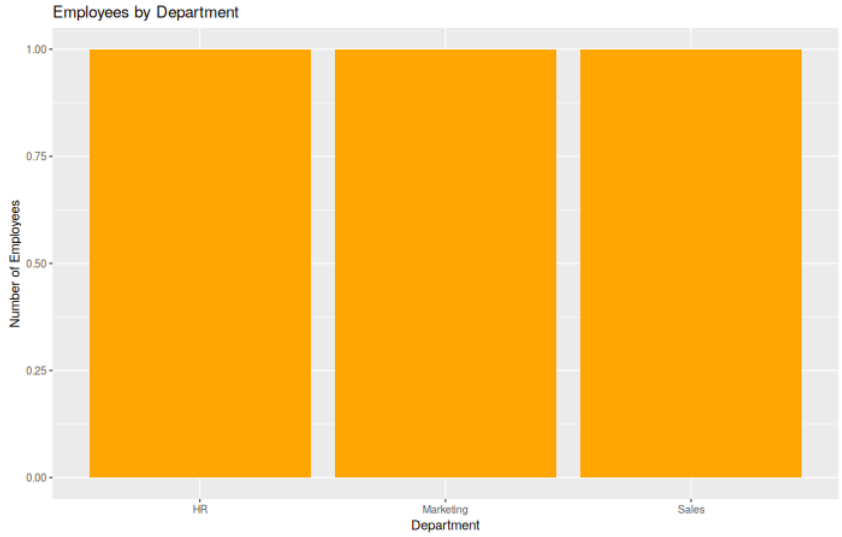
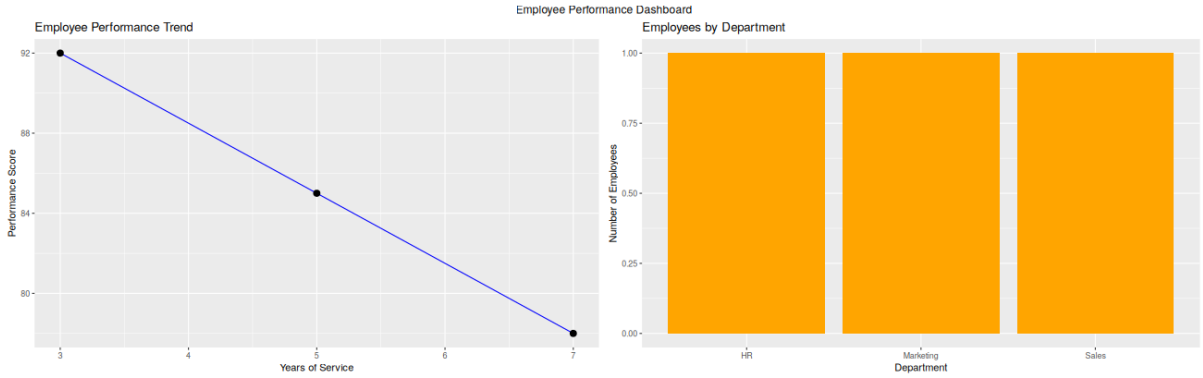


Figure 2: Question 2



EmployeeID	Department	YearsOfService	PerformanceScore
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Figure 3: Dashboard (Question 4)

Result

The line chart, bar chart, performance table and combined dashboard for employee performance were created successfully using R.